

Multilevel Methods on Finite Elements

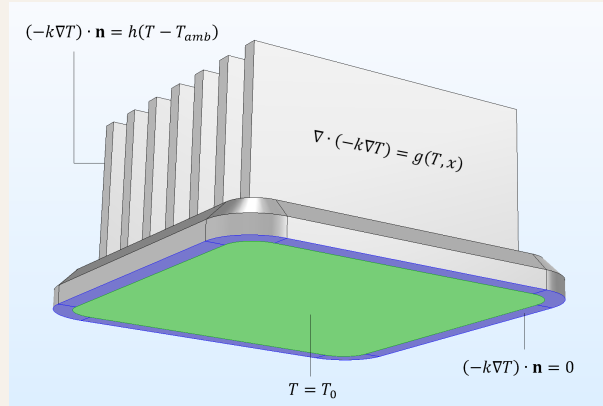
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Portland State University

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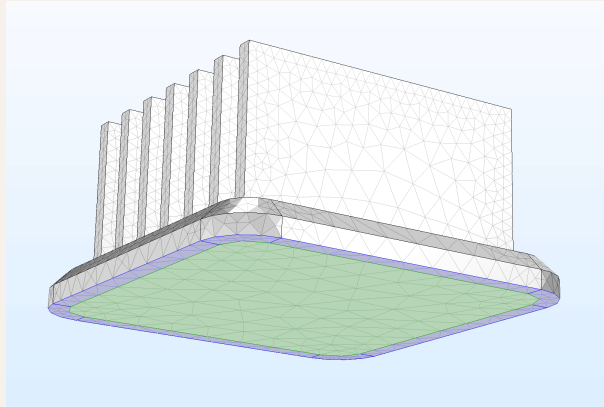
Modeling Physics

- ▶ Heat dissipating through a material.
- ▶ Problems include modeling how water flows from a source through porous materials.



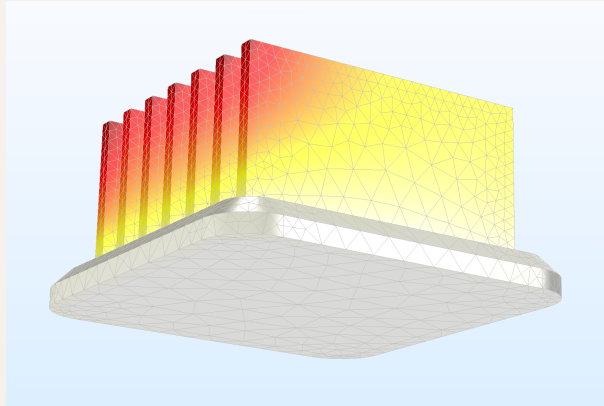
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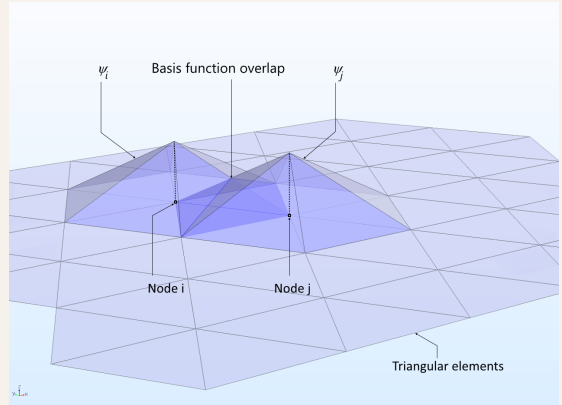
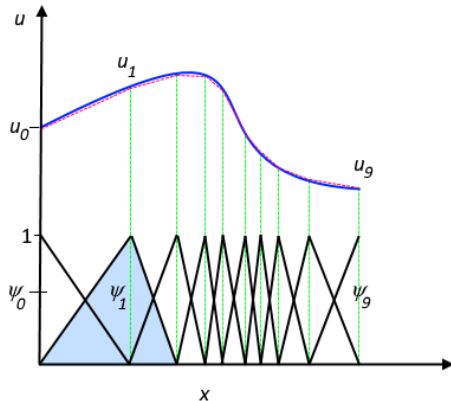
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PDEs and Finite Elements

To approximate the partial differential equation solutions, we use a basis of functions that behave in a manner we understand and that will simplify calculations.



Key Elements of Finite Elements

- 1 Start with a PDE
- 2 Weak form
- 3 Divergence theorem/integrate by parts
- 4 Discretize the domain
- 5 Represent the weak form as a discretization on the parts of the mesh
- 6 The operators become matrices
- 7 Solve with linear algebra
- 8 Use norms to see how good your approximation is to toy problems where you know the solution

Really Big Problems

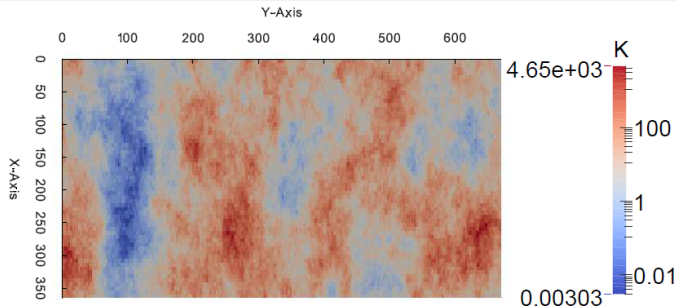


FIG. 2. SPE10 x -permeability for the top layer.

TABLE 2

Degrees of freedom, number of nonzeros, and complexities.

Problem	#elements	#faces	#DoFs	nnz	Arithmetic complexity	Operator complexity
60×220 (fine)	13200	53080	92680	607485	-	-
30×110	3317	13418	23371	152965	1.25252	1.2518
15×55	842	3446	5986	39327	1.06491	1.06474
6×22	149	634	1093	7351	1.01199	1.0121

Too big to run

Like Hillary said yesterday, some of these problems are absurdly large. We have to find ways to solve them when $x = A^{-1}b$

Netgen/NGSolve is a high performance multiphysics finite element software. It is widely used to analyze models from solid mechanics, fluid dynamics and electromagnetics. Due to its flexible Python interface new physical equations and solution algorithms can be implemented easily.



All in One

Seamless integration from geometric modeling, mesh generation, numerical simulation to visualization



Flexible

Mathematical description of variational formulation allows coupling of arbitrary physical models



Accurate

Cutting edge numerical techniques: high order, vectorial, mixed and discontinuous, Galerkin methods



Efficient

Robust preconditioners adapted to function spaces and differential equations



High Performance

Parallel compute core written in modern C++ combined with flexible steering by Python scripting

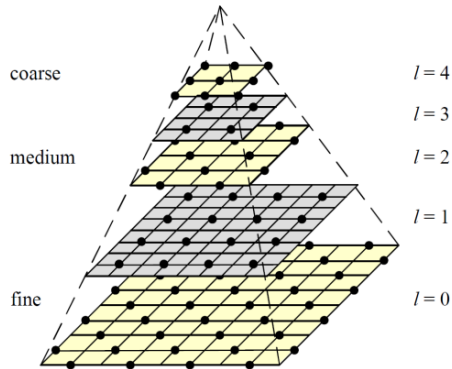


Open

Open source based on the LGPL license, extendable by C++ modules and Python packages



A Hierarchy of Finite Element Spaces



Setting up the spaces

At each level, we have the data storage, solving tools, and projection matrices available.

```
186     mesh: ng.Mesh
187     fes: object
188     a: BilinearForm
189     f: LinearForm
190     gfu: GridFunction
191     free_ids: np.ndarray
192     fixed_ids: np.ndarray
193     P: Optional[object] = None
194     PT: Optional[object] = None
195
```

Gauss Siedel Smoothing

Example

Two level method
Use Gauss Siedel to smooth the matrix, then

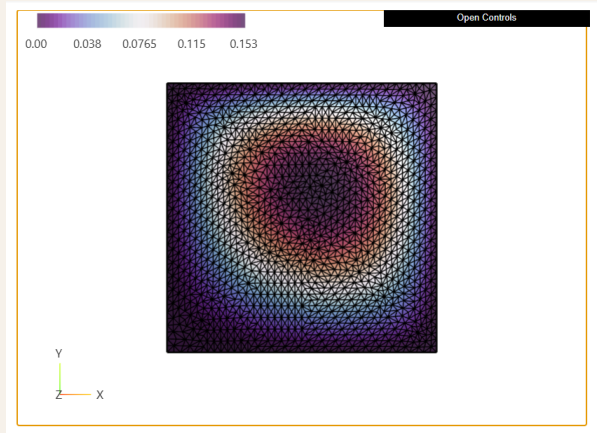
$$x_{k+1} = b - M^{-1}Ax_k \quad (1)$$

fine ndof=1961, coarse ndof=511,
 P shape 1961x511

$\|e\|_E$ (after section 1 pre-smooth):
1.790578e+00

$\|e\|_E$ after coarse correction:
1.126177e-01

$\|e\|_{L2}$ error: 9.820784e-04



Gauss Siedel Smoothing

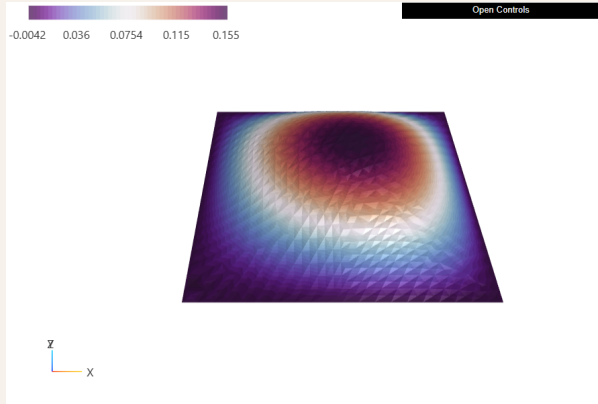
Example

Single-level smoother
fine ndof=1961, coarse ndof=511,
 P shape 1961x511

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Gauss Siedel Smoothing

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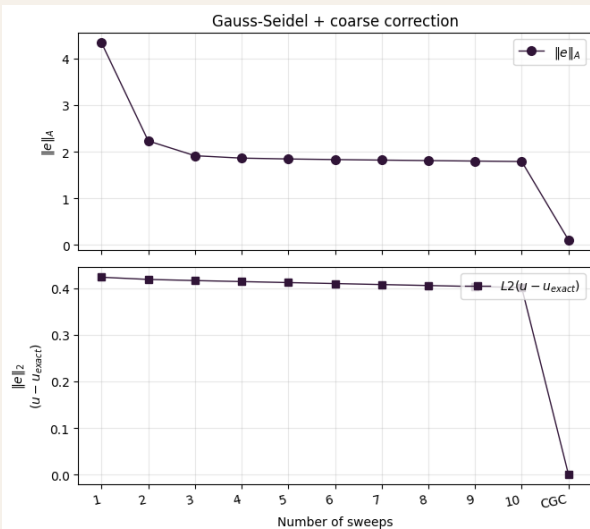
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Solving V-cycle with smoothing and coarse solve

start: $x \leftarrow x_0$

Level 2 (fine)
pre-smooth: relax $A_2 x = f$
residual $r_2 = f - A_2 x$
restrict $b_1 = P_2^\top r_2$

restrict
pass RHS
 $b_1 = P_2^\top r_2$

Level 1
pre-smooth: relax $A_1 e_1 = b_1$
residual $r_1 = b_1 - A_1 e_1$
restrict $b_0 = P_1^\top r_1$

restrict
pass RHS
 $b_0 = P_1^\top r_1$

Level 0 (coarse)
direct solve $e_0 = A_0^{-1} b_0$

prolongate
pass correction
 e_0

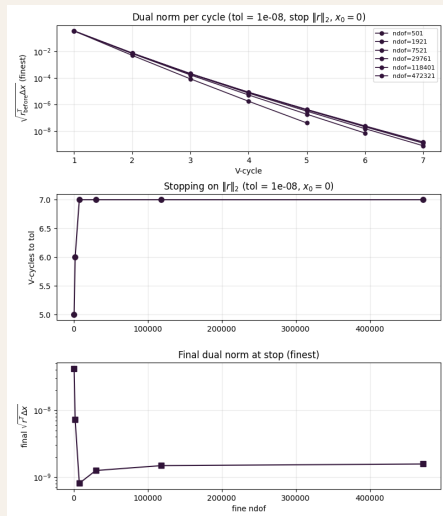
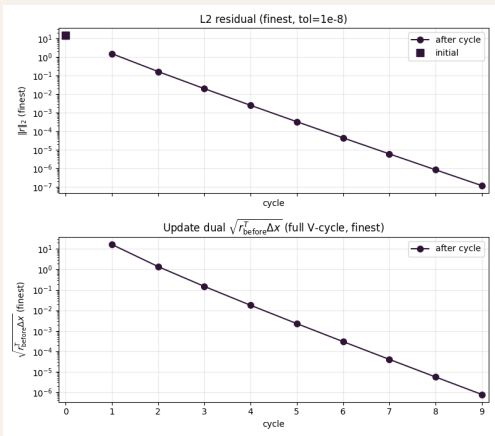
Level 1
update $e_1 += P_1 e_0$
post-smooth: relax $A_1 e_1 = b_1$

prolongate
pass correction
 e_1

Level 2 (fine)
update $x += P_2 e_1$
post-smooth: relax $A_2 x = f$

finish: updated x

Convergence of v-cycle



Now we use it as a preconditioner

Success! We made it so that we run PCG and when it asks for a preconditioner, it gets our two-level method PCG (NGSolve CG + your V-cycle): 1684 iterations

Conclusion

- ▶ We demonstrated the smoothing effects of Gauss Siedel on highly oscillatory components.
- ▶ We implemented a two-level method with GS smoothing and a coarse direct solve.
- ▶ We built a framework from which future multilevel test on NGSolve finite element spaces may be performed and tested it to demonstrate convergence.
- ▶ We implemented the method above as the preconditioner for a preconditioned conjugate gradient solver.

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- ▶ P. S. Vassilevski, "Multilevel Block Factorization Preconditioners, Matrix-based Analysis and Algorithms for Solving Finite Element Equations", Springer, New York, 2008. 514 p.
- ▶ J. S. Owall, Numerical Mathematics. Philadelphia, PA: SIAM, 2024. DOI: 10.1137/1.9781611978070.
- ▶ J. N. Reddy, N. K. Anand, and P. Roy, Finite Element and Finite Volume Methods for Heat Transfer and Fluid Dynamics. Cambridge University Press, Cambridge, UK, 2023. DOI: 10.1017/9781009275453.
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- ▶ Some finite element images came from <https://www.comsol.com/multiphysics/fea-software>